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ALEX MENDELSON

Evolutionary Ecologist, Science Communicator

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Education

MS Biology | California State University, Northridge — Expected June 2026

BS Organismal Biology | California State University, Long Beach — 2022

Research Projects

terHorst Evolutionary Ecology Lab | CSU Northridge

- Evolution-mediated indirect effects of climate on interspecific competition — 2025–present
  Data component: Custom parameterized growth modeling, structural equation modeling, multi-factorial ANOVA
  Publication: manuscript in preparation; Chapter 2 of MS thesis
- Complementarity of intraspecific diversity on rotifer competitive ability — 2025–present
- Spatial distribution of diverse temperature responses in a Bdelloid rotifer — 2023–present
  Data component: Non-parametric growth modeling, linear mixed model analysis
  Publication: manuscript in preparation; Chapter 1 of MS Thesis
- Automating measurement of rotifer body size using computer vision models — 2025–2025
  Data component: Built a semi-automated image analysis pipeline (Python, OpenCV, Fiji, Ilastik) to extract measurements from ~6,000 images

Pernet Larval Ecology Lab | CSU Long Beach

- Larval settlement behavior in an aggregation-forming annelid — 2022–2025
  Publication: **Mendelson et al., 2025** *(see below)*
  Data component: Linear mixed model analysis
- Phenotypic plasticity in larval morphology of an Echinoid — 2021–2022
- Automated algal feeding system for rearing marine invertebrate larvae — 2021–2021
  Data component: regularly cleaned and organized over 160,000 data points for analysis during instrument testing

Publications

Mendelson, A. A., K. Nuon, B. Pernet. (2025). Gregarious and non-gregarious larval settlement in the aggregation-forming annelid *Ficopomatus enigmaticus*. *Biol. Bull.* 248:1-10. https://doi.org/10.1086/740316

Data Skills

Statistical Analyses

- Structural equation modeling
- Multivariate (PCA, NMDS)
- Random effects (LMM)
- Analysis of variance: (ANOVA, ANCOVA, MANOVA, MANCOVA)
- Generalized linear models
- Custom parametrized models

Languages/Software

- R
- Python
- Git & GitHub
- ImageJ (incl. Fiji)
- Ilastik
- Adobe Creative Suite
- MS Office Suite

Data Visualization

- Interactive data visualization (R Shiny)
- ggplot2 (R)
- Publication-quality figure design
- Figure refinement and layout (Adobe Illustrator)

Science Communication

Graduate Assistant, *Biometry (BIOL 552)* 2025–2025
Advanced analysis of biological data | CSU Northridge

Peer Learning Facilitator, *Principles of Biology I Lecture (BIOL 106)* 2025–present
Guided office hours for intro biology lecture students | CSU Northridge

Teaching Associate, *Principles of Biology I/II Lab (BIOL 106/7L)* 2023–present
Intro biology lab courses (21 sections) | CSU Northridge

Education Intern, *Bolsa Chica Conservancy* 2022–2023
Tour guide of coastal wetlands, K-12 and adult | Bolsa Chica Conservancy

Service & Community

Tree Canopy Assessment Research Project | US Forest Service, CSUN Urban Forestry 2026
Field survey of trees in the fire-affected neighborhoods of Altadena, CA

President | Behavior, Ecology, and Evolution Research Club, CSUN 2024–2025
Facilitated weekly meetings, where grads and undergrads discuss relevant literature

Program Assistant | Piece by Piece 2022–2023
Assisted with free art workshops for currently and recently unhoused community members in Skid Row and South LA

Executive Producer | Agewell Long Beach, Heart of Ida 2020–2021
Filmed and produced a wellness public access TV program targeting the aging population in Long Beach, CA during the COVID-19 pandemic

Invited Talks

What actually happens when your computer "models"? 2026
Guest lecturer, CSUN Biol 551 (Computer Modeling)

Local butterfly ecology 2026
Creative Habitat: Ecology Meets Mosaic, Crafted at the Port of LA

Presentations

** denotes presenter*

Mendelson, A. A*., C. P. terHorst, 2026. Adaptation to climate change and its indirect effects on rotifer ecology. Southern California Academy of Sciences, Anaheim, CA [**talk**]; CSUNposium, Northridge, CA [**talk**]

Mendelson, A. A*., C. P. terHorst. 2025. The search for indirect ecological effect of evolution in pitcher plant rotifers. Western Society of Naturalists, San Diego, CA [**talk**]

Mendelson, A. A*., C. P. terHorst. 2025. Does response to increased temperature differ among clones of pitcher plant rotifers? Evolution, Athens, GA [**poster**]

Mendelson, A. A*., C. P. terHorst. P. 2024. Different effects of temperature on pitcher plant rotifer clones. Ecological Society of America, Long Beach, CA [**poster**]

Mendelson, A. A*., C. P. terHorst. 2024. Ecological effect of evolution in pitcher plant rotifers as a response to temperature stress. CSUNposium, Northridge, CA [**poster**]

Mendelson, A. A*., K. Nuon, B. Pernet. 2022, 2023. It's all a big mistake: larval errors and the formation of new aggregations of the serpulid annelid *Ficopomatus enigmaticus*. Western Society of Naturalists, Monterey, CA [**talk**]; Southern California Academy of Sciences, Santa Barbara, CA [**talk**]; California Estuarine Research Society, Costa Mesa, CA [**poster**]; Western Society of Naturalists, Oxnard, CA [**poster**]